Exercise-5

Loops

(a) "for" Loop in C

Objective: Learn the usage of the for loop

```
#include<stdio.h>
int main()
{
   int i, sum = 0;
       for(i = 1; i <= 5; i++)
   {
      printf("%d\n",i);
              sum = sum + i;
   }
       printf("Sum = %d", sum);
   return 0;
}
```

O/P

1

2

3

4

5

Sum = 15

(b) Sum of the digits of a 5-digit number

Objective: Learn the usage of while loop and usage of operators - % and /.

```
#include<stdio.h>
int main()
{
   int num;
   printf("Enter a 5-digit number: ");
   scanf("%d", &num);
   int sum = 0;
   while(num != 0)
   {
      sum += num % 10;
      num = num/10;
}
printf("Digit sum: %d", sum);
return 0;
}
```

O/P

Enter a 5-digit number: 12345

Digit sum: 15

(c) Given number is a prime or not. (Also Prime numbers between a given range.

```
//prime or not
#include<stdio.h>
int main()
{
int n, i, count = 0;
scanf("%d", &n);
for(i = 1; i <= n; i++)  // check divisibility from 1 to n
{
if(n % i == 0)
{
count++;
// if remainder is 0, i is a factor
// count number of factors
}
}
if(count == 2)
{
// prime numbers have only 2 factors (1 and itself)
printf("%d is Prime number",n);
}
else
{
```

```
}
printf("%d is NOT a Prime number",n);
return 0;
}
```

<u>O/P</u>

6

6 is NOT a Prime number

<u>Prime numbers between a given range</u>

```
#include<stdio.h>
int main()
{
   int start, end, i, j, count;
   scanf("%d%d", &start, &end);
   printf("Prime numbers between %d and %d are:\n", start,
end);
   for(i = start; i <= end; i++)    // check each number
   {
      count = 0;
      for(j = 1; j <= i; j++)     // count factors of i
      {
         if(i % j == 0)
         {
            count++;
```

```
            }
        }
        if(count == 2)              // prime if exactly 2 factors
        {
            printf("%d\n", i);
        }
    }
    return 0;
}
```

<u>O/P</u>

2 10

Prime numbers between 2 and 10 are:

2

3

5

7

(d) <u>Armstrong Number or not.</u>

```
#include<stdio.h>
int main()
{
 int n, r, sum=0, t;
 scanf("%d",&n); //Enter 3digits number
 t=n;
 printf("\n the given nuber=%d",n);
```

```
 while(n>0)
{
r=n%10;
sum=sum+(r*r*r);
n=n/10;
}
if(t==sum)
{
	printf("\n%d is Armstrong Number",t);
}
else
{
	printf("\n%d is Not Armstrong Number",t);
}
return 0;
}
```

O/P

153

the given nuber=153

153 is Armstrong Number

(e) Palindrome or not

```
// Program to check whether a number is palindrome or not
#include<stdio.h>
int main()
{
   int n, r, t, rev = 0;
   scanf("%d", &n); //Enter any Number
       t = n;  // Store original number
   printf("The given number = %d", n);
       // Reverse the number
   while(n > 0)
   {
      r = n % 10;         // Get last digit
      rev = rev * 10 + r; // Build reverse number
      n = n / 10;         // Remove last digit
   }
 printf("\nThe reverse value = %d", rev);
 // Check palindrome
   if(t == rev)
   {
      printf("\n%d is a palindrome", t);
   }
   else
   {
```

```c
        printf("\n%d is not a palindrome", t);
    }
 return 0;
}
```

O/P

232

The given number = 232

The reverse value = 232

232 is a palindrome

(f) Objective: Print a pattern of numbers using Loops.

```c
//using nested for loop to print Square pattern
#include<stdio.h>
int main()
{
   int i, j, n;
   scanf("%d",&n);

   for(i = 1; i <= n; i++)       // outer loop rows
   {
      for(j = 1; j <= n; j++)   // inner loop  columns
      {
         printf("%d ", j);
      }
      printf("\n");             // move to next line after each row
```

```
    }

   return 0;
}
```

O/P

3

1 2 3

1 2 3

1 2 3

(g) Construct a Pyramid pattern.

```
//using nested for loop to print half pyramid
#include<stdio.h>
int main()
{
 int i,j;
 for(i=1;i<=5;i++)
 {
  for(j=1;j<=i;j++)
  {
   printf("%d ",j);
}
printf("\n");
}
```

return 0;

}

O/P

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5